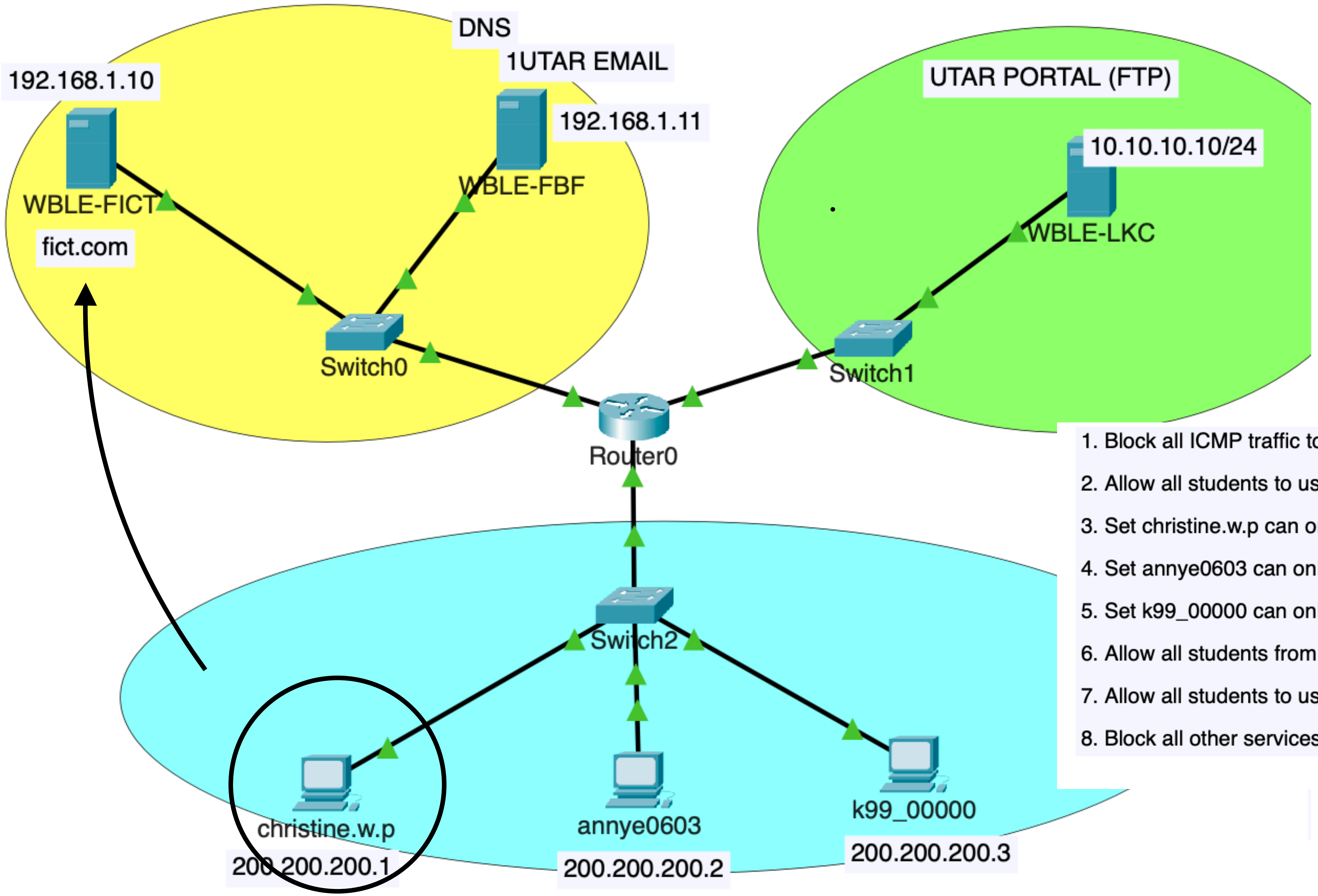
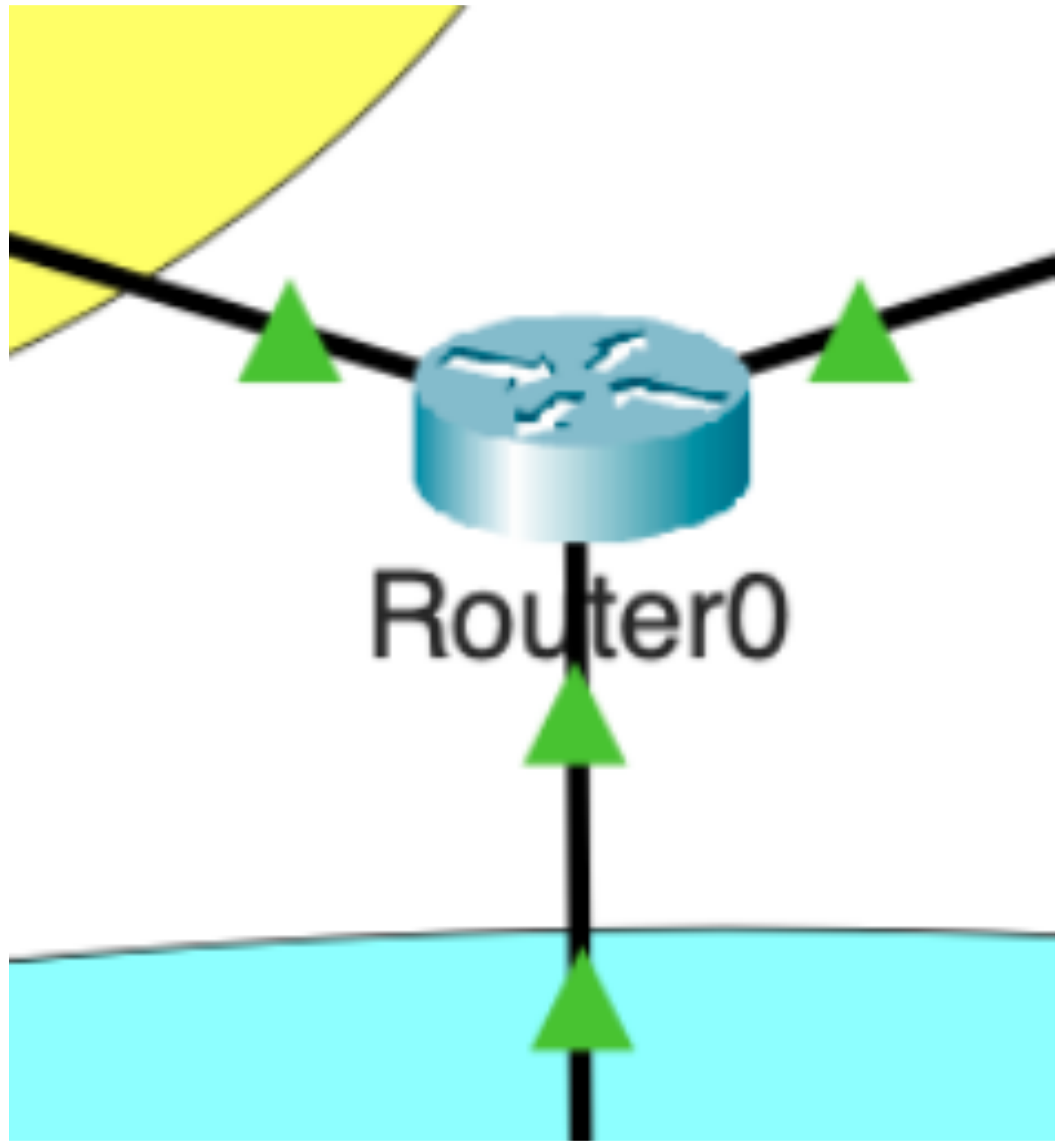
A day in the life of UTAR_ADMIN

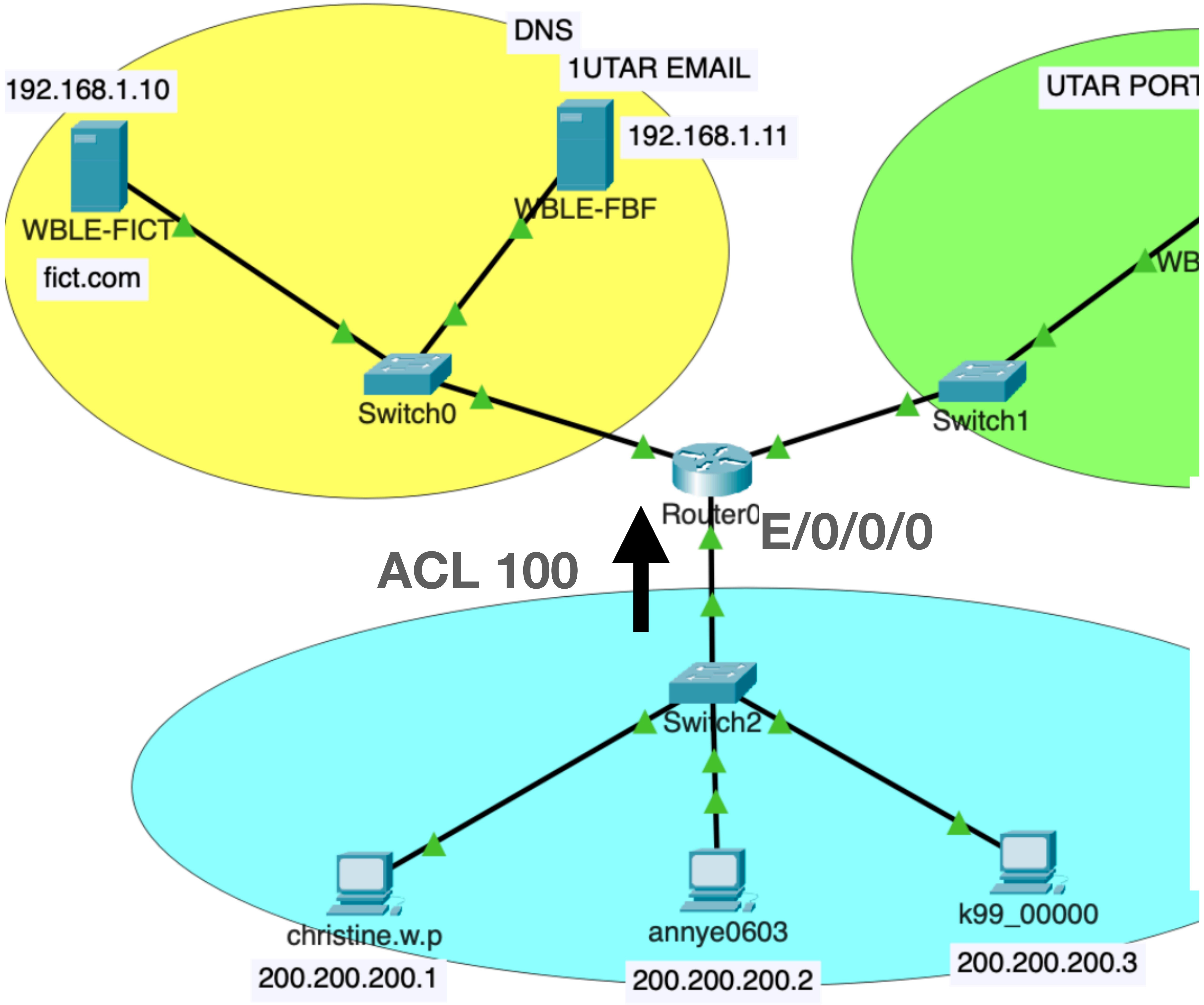
WBLE use port 80
DNS use port tcp/53 and udp/53



1. Block all ICMP traffic to prevent DDOS
2. Allow all students to use DNS services
3. Set christine.w.p can only go to WBLE-FICT but not WBLE-FBF and WBLE-LKC
4. Set annye0603 can only go to WBLE-FBF but not WBLE-FICT and WBLE-LKC
5. Set k99_00000 can only go to WBLE-LKC but not WBLE-FICT and WBLE-FBF
6. Allow all students from anywhere to use 1UTAR email service
7. Allow all students to use UTAR Portal (FTP) from anywhere
8. Block all other services that are not listed in UTAR policy like douyin and facebook app

ACL 100 is already placed at e0/0/0 IN





Create ACL 100

1. Block all ICMP traffic to prevent DDOS
2. Allow all students to use DNS services
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1. Block ICMP

PING command uses ICMP

- Ping is used to check if host A can talk to host B

- On Windows, the sender sends 4 ICMP requests to the destination

  - If the destination is reachable, it will reply 4x ICMP replies
  - If destination unreachable, the router will reply error message to sender

- On Linux/MAC OS, the ping command is unlimited. It will not stop until you press CTRL + C to stop it

- If we use many Linux machines and ping to a same target (SPAM ping), very likely the destination will crash due to too many ICMP packets to process. This is a common method used by attackers to DDOS targets.

- In UTAR, *utar_admin* must block all ICMP so that students cannot mass PING WBLE server

1. Block ICMP

PING command uses ICMP

- To block PING, we can use extended ACL

```
access-list [ID:100+] [permit/deny] [protocol] [sourceIP] [destIP] eq [port/name]
```

`access-list 100 deny icmp any any`

```
[protocol] = icmp
[sourceIP] = any
[destIP] = any
```

2. Allow students to use DNS

So that everyone can browse websites

- DNS is required so that students can visit any websites
- We use whitelist since we want to allow DNS for all students

```
access-list [ID:100+] [permit/deny] [protocol] [sourceIP] [destIP] eq [port/name]
```

```
Access-list 100 permit udp any host 192.168.1.11 eq domain

Access-list 100 permit udp any host 192.168.1.11 eq 53
```

```
[Protocol] = udp
[sourceIP] = any
[destIP] = host 192.168.1.11
[Port] = 53 or DNS
```

3. Allow Christine to go WBLE FICT server only

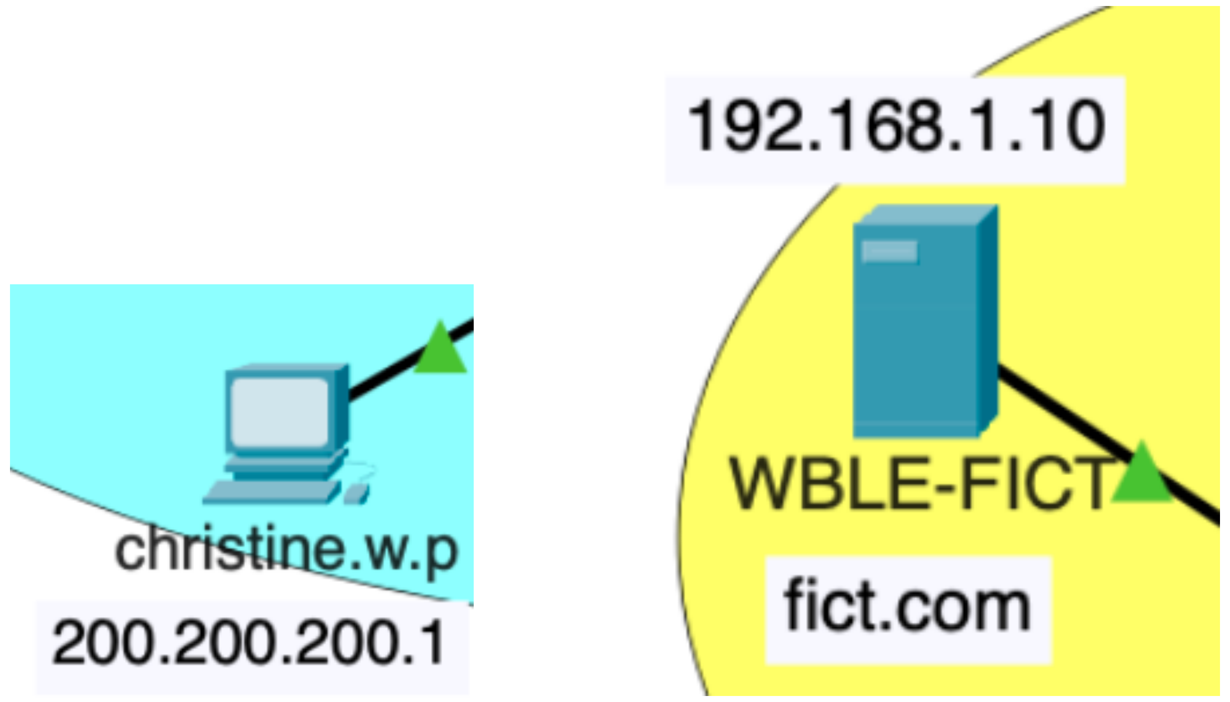
Christine cannot go to WBLE-FBF and WBLE-LKC

- WBLE is a web server. Web server use http or https protocol, runs on port 80/ port 8080/ port 443.
- In this example, we use port http/80.
- We use whitelist approach, as we know which server to allow, but we do not know all other servers to be blocked.

```
access-list [ID:100+] [permit/deny] [protocol] [sourceIP] [destIP] eq [port/name]
```

```
Access-list 100 permit tcp host 200.200.200.1 host 192.168.1.10 eq 80
```

```
Protocol = tcp
sourceIP = host 200.200.200.1
destIP = host 192.168.1.10
Port = 80
```



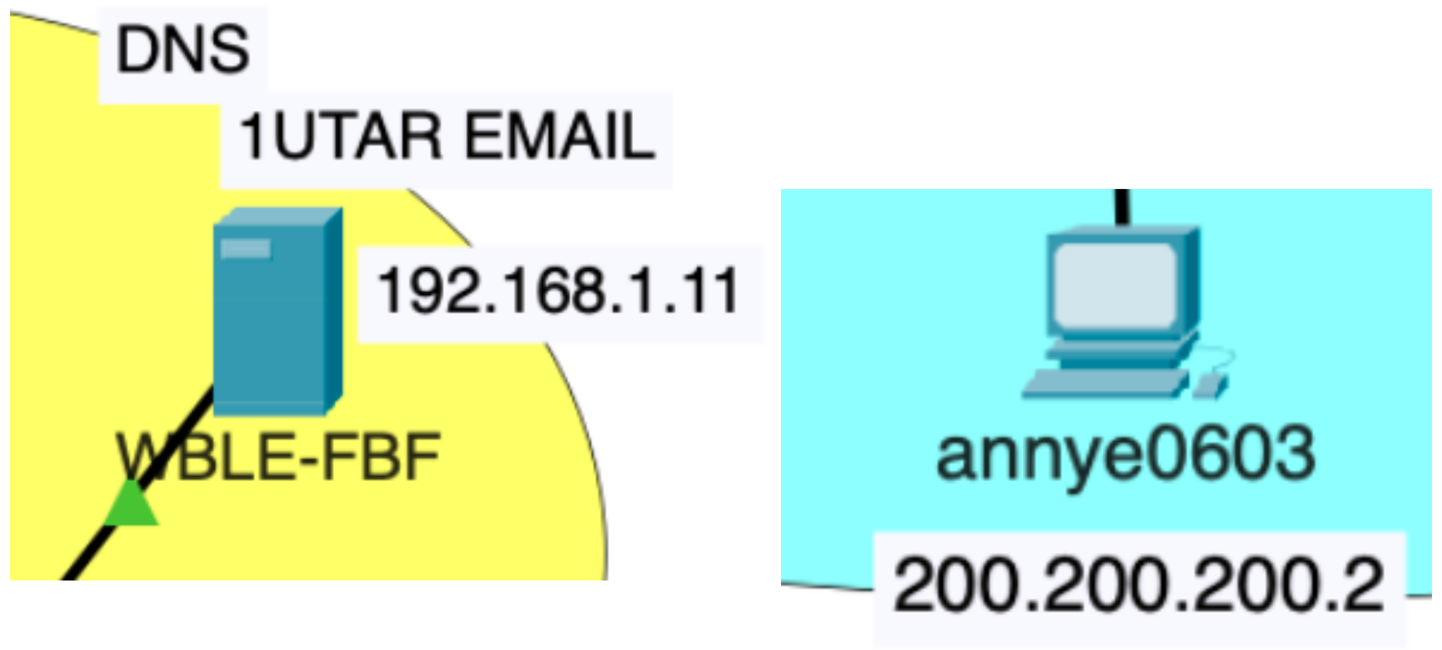
4. Allow annye to go WBLE FBF server only

Annye cannot go to WBLE-FICT and WBLE-LKC

```
access-list [ID:100+] [permit/deny] [protocol] [sourceIP] [destIP] eq [port/name]
```

```
Access-list 100 permit tcp host 200.200.200.2 host 192.168.1.11 eq 80
```

```
Protocol = tcp
sourceIP = host 200.200.200.2
destIP = host 192.168.1.11
Port = 80
```



5. Allow k99 to go WBLE LKC server only

K99 cannot go to WBLE-FICT and WBLE-FBF

```
access-list [ID:100+] [permit/deny] [protocol] [sourceIP] [destIP] eq [port/name]
```

```
Access-list 100 permit tcp host 200.200.200.3 host 10.10.10.10 eq 80
```

```
Protocol = tcp
sourceIP = host 200.200.200.3
destIP = host 10.10.10.10
Port = 80
```

6. Allow all students to use 1utar email from anywhere

- utar_admin must setup emails for all students
- Email use two protocols: (a) SMTP to send and (b) POP3 to receive

Since we ALLOW for all students, from any network, we use ANY keywords for sourceIP

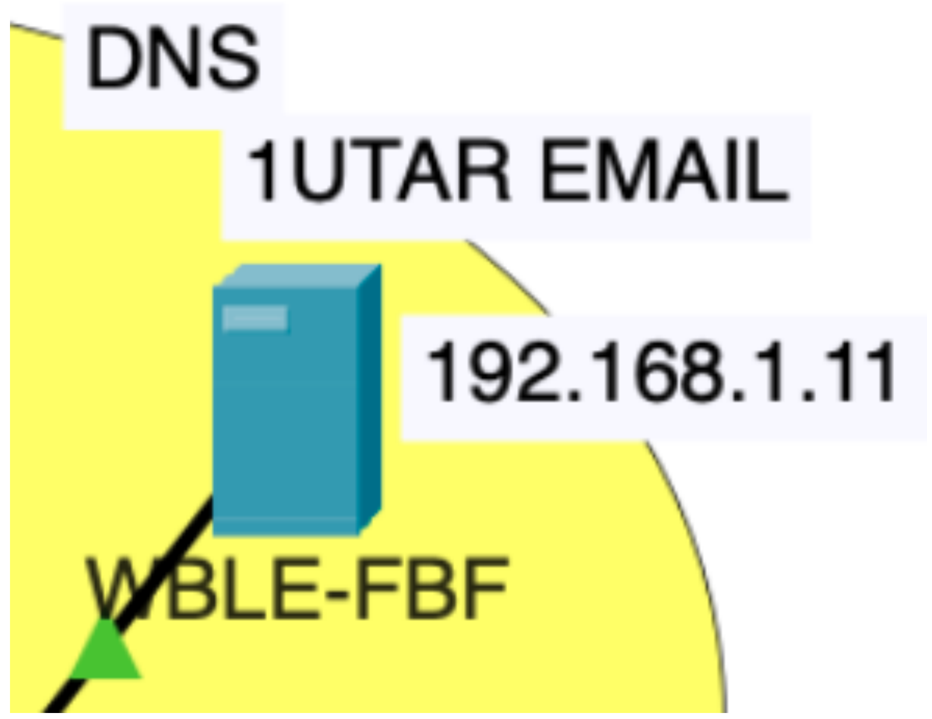
```
access-list [ID:100+] [permit/deny] [protocol] [sourceIP] [destIP] eq [port/name]
```

```
Access-list 100 permit tcp any host 192.168.1.11 eq smtp
Access-list 100 permit tcp any host 192.168.1.11 eq pop3
```

christine@1utar.my

anny@1utar.my

```
Protocol = tcp
sourceIP = any
destIP = host 192.168.1.11
Port = smtp, pop3
```



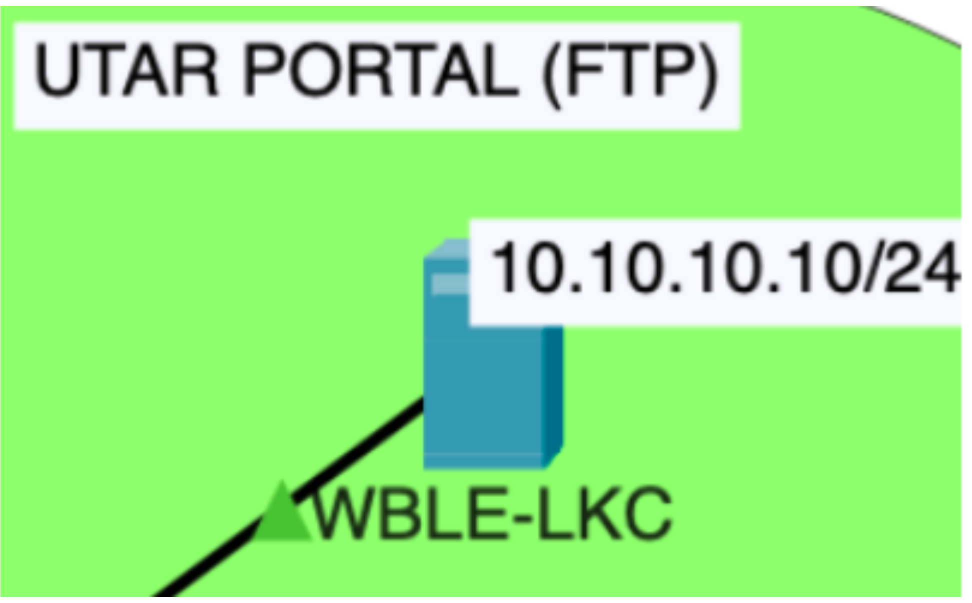
7. Allow all students to use FTP from anywhere

- utar_admin must setup FTP for all students

```
access-list [ID:100+] [permit/deny] [protocol] [sourceIP] [destIP] eq [port/name]

Access-list 100 permit tcp any host 10.10.10.10 eq ftp
```

```
Protocol = tcp
sourceIP = any
destIP = host 10.10.10.10
Port = ftp
```



8. Block everything else

- utar_admin must block everything else not explicitly allowed (in the whitelist).
- After finish setting all rules, utar_admin can use the ‘show run’ or ‘show access-lists’ command to check the setting.

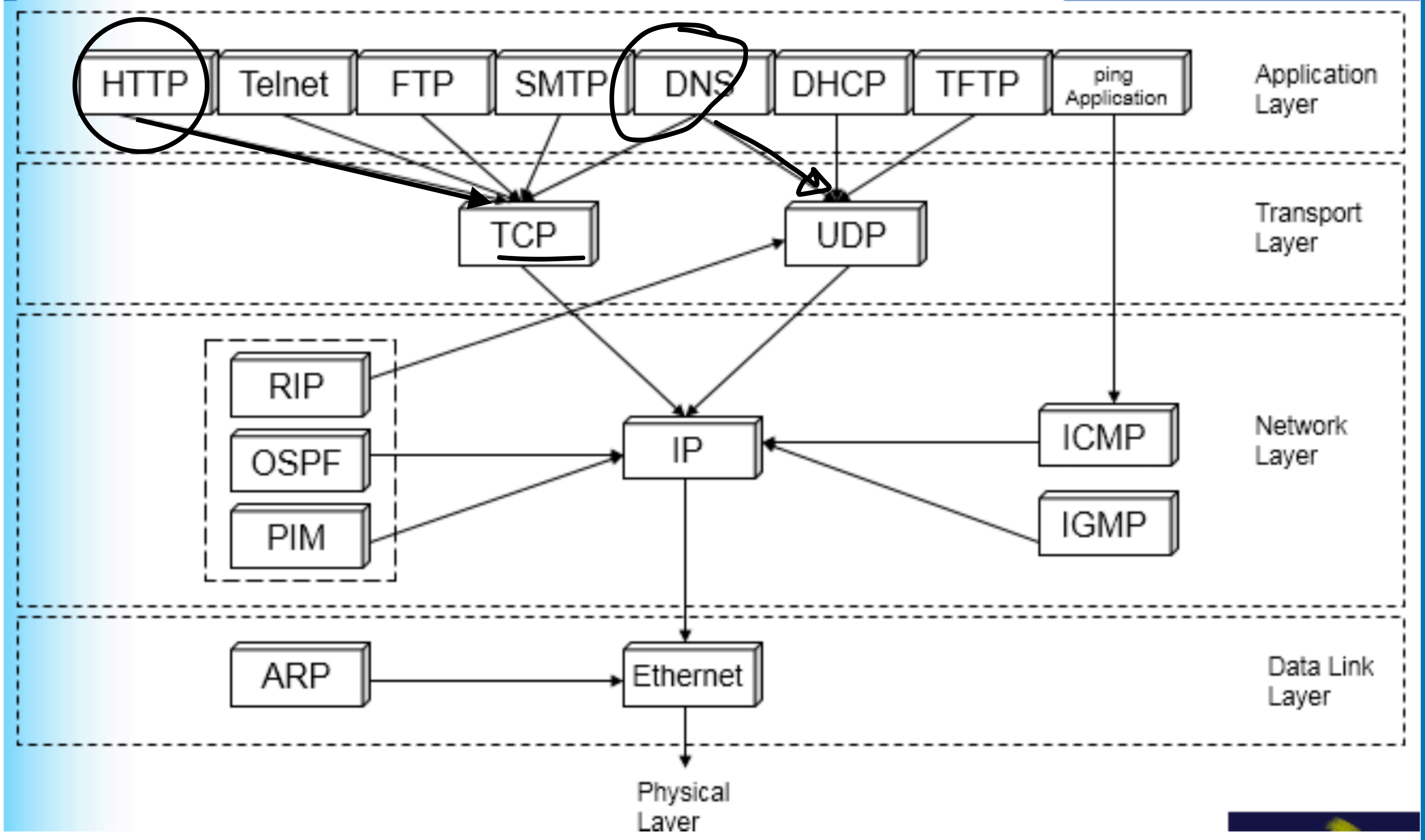
This command is optional. Since router has the default DENY ALL statement in ACL. All traffic not allowed earlier are already blocked by default

```
access-list [ID:100+] [permit/deny] [protocol] [sourceIP] [destIP] eq [port/name]
```

```
Access-list 100 deny any any
```

The bible of the Internet

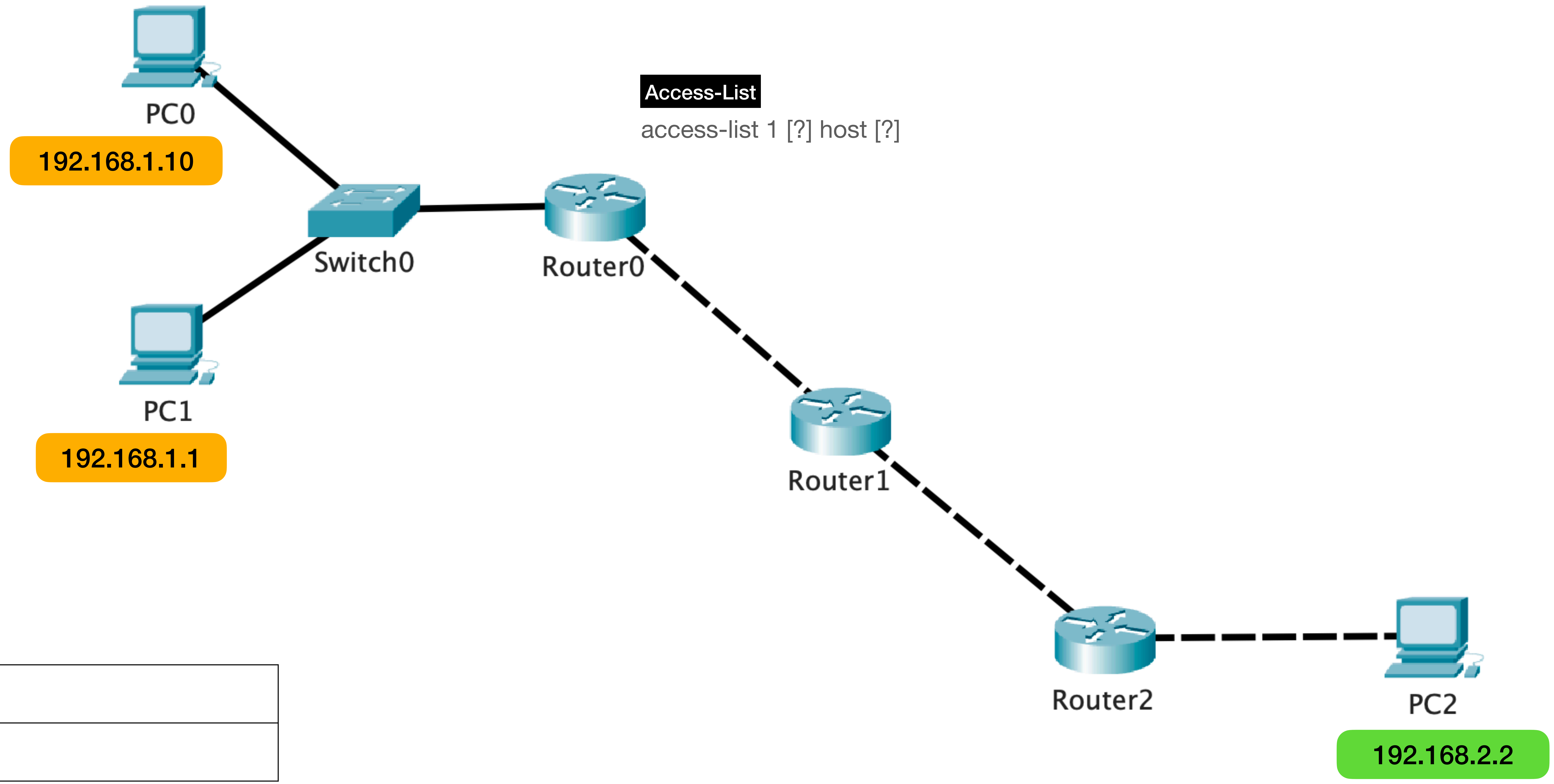
TCP/IP Protocol Stack



Extra Examples

Access-List

access-list 1 [?] host [?]



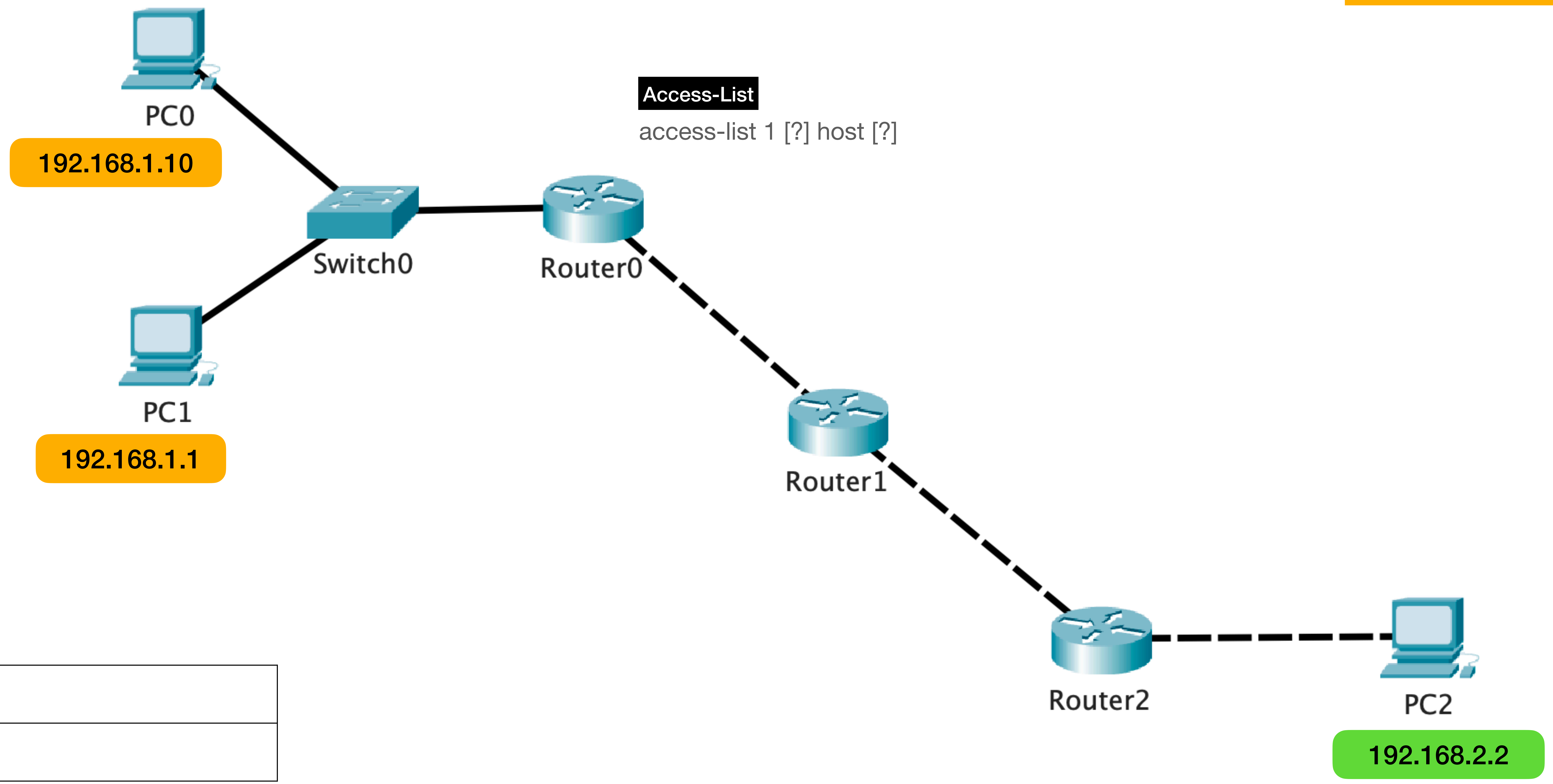
SrcIP	
DestIP	

Standard ACL Syntax: access-list [ID] [permit/deny] [source IP]

Block PC0 to PC2
Allow PC1 to PC2

Access-List

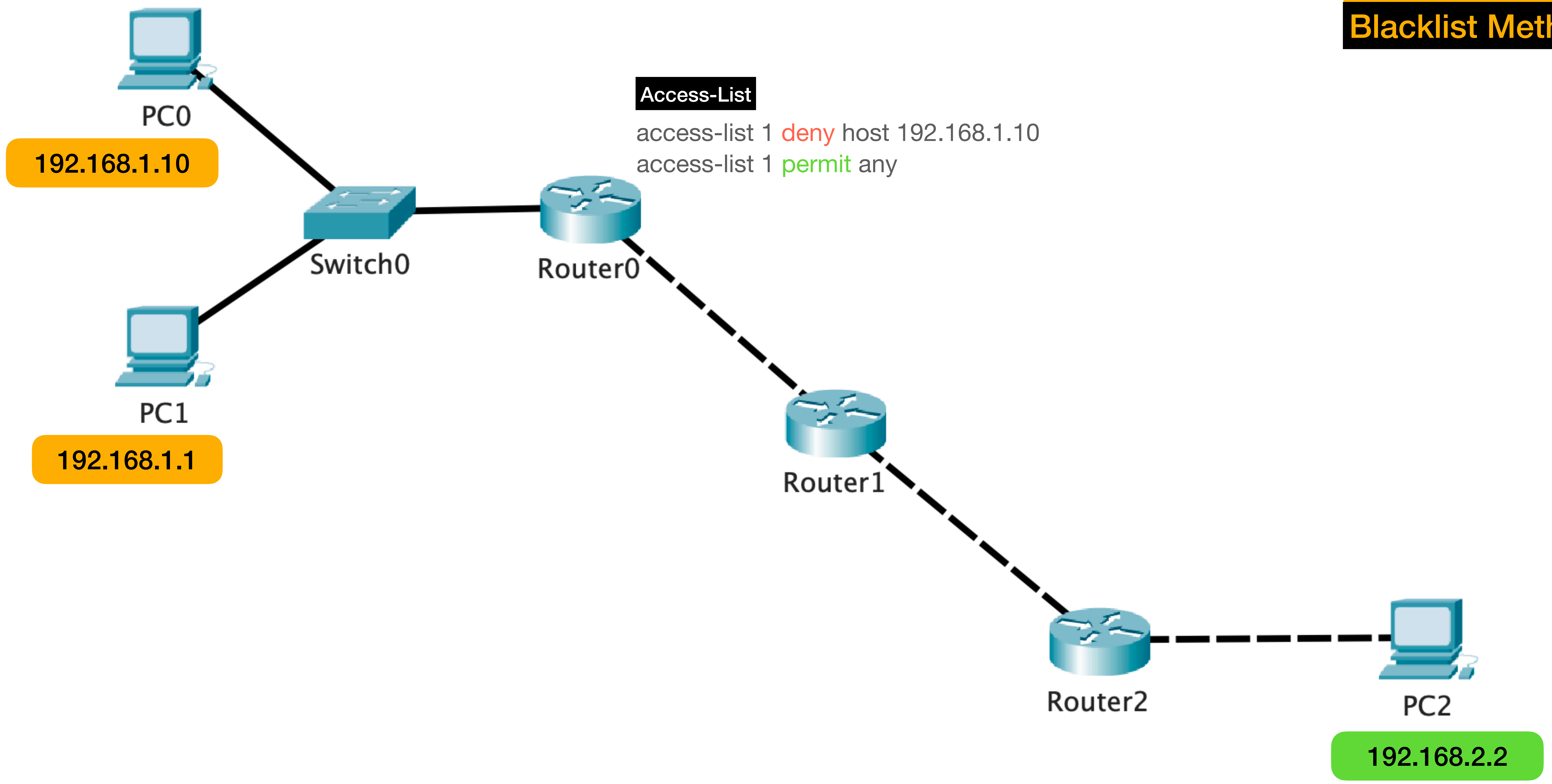
access-list 1 [?] host [?]



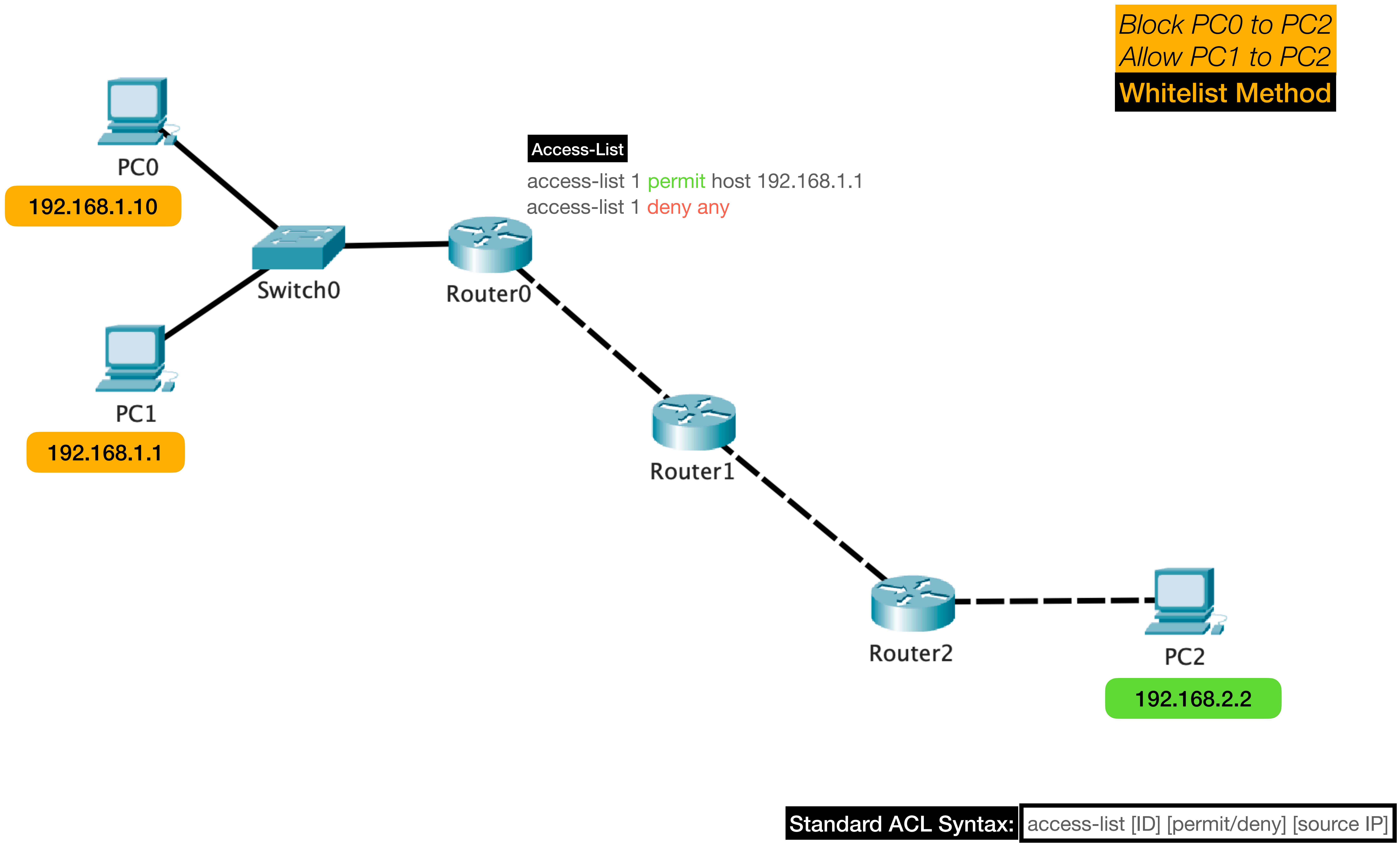
SrcIP	
DestIP	

Standard ACL Syntax: access-list [ID] [permit/deny] [source IP]

Block PC0 to PC2
Allow PC1 to PC2
Blacklist Method



Standard ACL Syntax: access-list [ID] [permit/deny] [source IP]


Block PC0 to PC2
Allow PC1 to PC2
Whitelist Method
PC0
192.168.1.10
Switch0
PC1
192.168.1.1
Access-List
access-list 1 permit host 192.168.1.1
access-list 1 deny any
Router0
Router1
Router2
PC2
192.168.2.2
Standard ACL Syntax:
access-list [ID] [permit/deny] [source IP]

Block PC0 to PC2
Allow PC1 to PC2

Placement

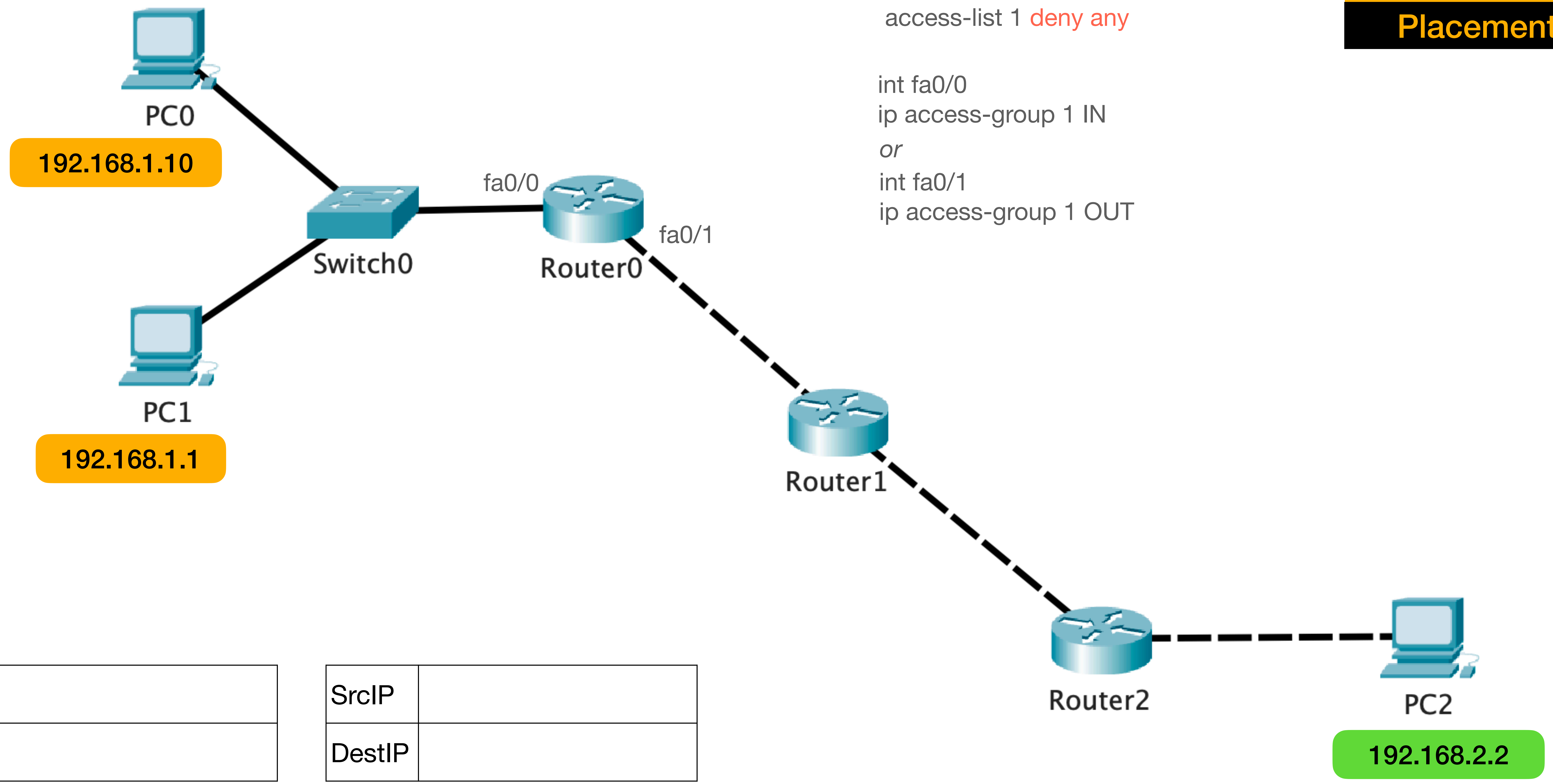
Access-List (R0)

```
access-list 1 permit host 192.168.1.1
access-list 1 deny any
```

```
int fa0/0
ip access-group 1 IN
```

or

```
int fa0/1
ip access-group 1 OUT
```



SrcIP	
DestIP	

PC1->PC2

SrcIP	
DestIP	

PC2->PC1

Standard ACL Syntax: ip access-group <ID> <IN/OUT>